

Name: AMRITANSU SINGH AMRITANSU SINGH | Reg. No: RA2511003011028

Unit 4 : Session 11 : SLO 2 | Converting a TCP Client to UDP

I. Key Changes (TCP client → UDP client)

1. Socket type: `socket.SOCK_STREAM` → `socket.SOCK_DGRAM`.
2. No connection setup: remove `connect()` → UDP is connectionless.
3. Sending: `send()/sendall()` → `sendto(data, (host, port))` → every datagram carries its destination.
4. Receiving: `recv()` → `recvfrom(bufsize)` → returns (data, address) because the sender must be identified.
5. No `listen()/accept()` on the peer; a UDP "server" just `bind()`s and calls `recvfrom()`.
6. Error handling: no automatic retransmission → timeouts and retries must be implemented by the application.

II. Example → UDP Client

- `import socket`
- `s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)`
- `s.sendto(b"Hello", ("localhost", 5000))`
- `data, addr = s.recvfrom(1024)`
- `print(data); s.close()`